

**ASSESSMENT TOOL – 2**  
**SCENARIO BASED ASSESMENT**

**Course:** C++ Object Oriented Programming

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**Hospital Consultation Charge System**

```
#include <iostream>

using namespace std;

float calculateCharge(float fee, float serviceCharge = 100);
float calculateCharge(float fee, int specialist, float serviceCharge = 100);
void updateBill(float &bill);

int main()
{
    int choice;

    float fee, totalBill;

    cout << "Hospital Consultation Charge System\n";

    cout << "1. General Consultation\n";
    cout << "2. Specialist Consultation\n";

    cout << "Enter your choice: ";

    cin >> choice;

    cout << "Enter Consultation Fee: ";

    cin >> fee;

    if (choice == 1)
    {
        totalBill = calculateCharge(fee);

        cout << "\nPatient Type : General\n";

    }

    else if (choice == 2)
    {
        totalBill = calculateCharge(fee, 1);

        cout << "\nPatient Type : Specialist\n";

    }
}
```

```

else
{
    cout << "Invalid Choice";

    return 0;
}

updateBill(totalBill);

cout << "Total Consultation Charge = " << totalBill << endl;

return 0;
}

float calculateCharge(float fee, float serviceCharge)
{
    return fee + serviceCharge;
}

float calculateCharge(float fee, int specialist, float serviceCharge)
{
    return fee + 500 + serviceCharge;
}

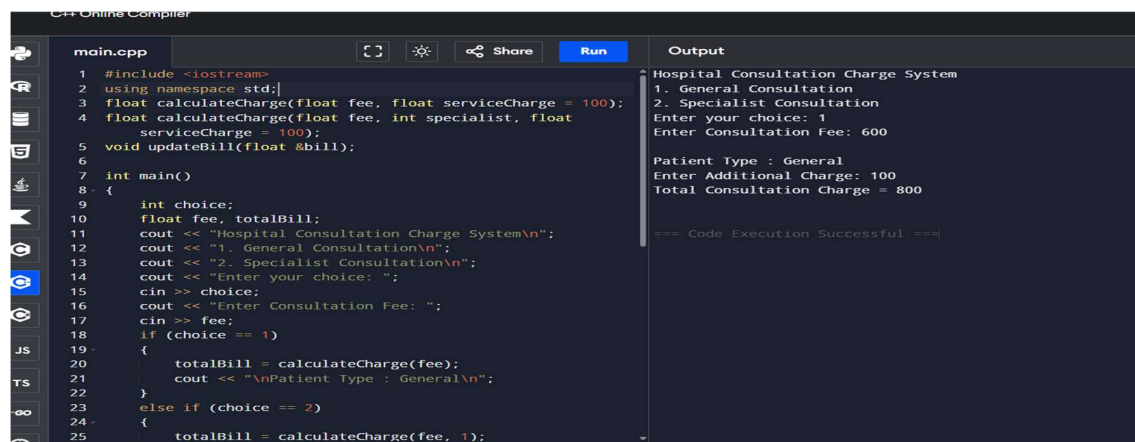
void updateBill(float &bill)
{
    float extraCharge;

    cout << "Enter Additional Charge: ";

    cin >> extraCharge;

    bill = bill + extraCharge;
}

```



The screenshot shows a C++ Online Compiler interface. On the left, there's a sidebar with icons for file management and a list of files including 'main.cpp'. The main editor area displays the C++ code from the previous block. On the right, the 'Output' panel shows the program's execution results. The output text is as follows:

```

Hospital Consultation Charge System
1. General Consultation
2. Specialist Consultation
Enter your choice: 1
Enter Consultation Fee: 600

Patient Type : General
Enter Additional Charge: 100
Total Consultation Charge = 800

=== Code Execution Successful ===

```

## **Fitness Centre Membership System**

```
#include <iostream>

#include <string>

using namespace std;

class Member
{
private:
    int memberID;
    string name, planType;
    float fee;
public:
    static int activeMembers;
    void assignPlan();
    void calculateFee();
    void display();
    static void showActiveMembers() {
        cout << "\nTotal Active Members: " << activeMembers;
    }
};

int Member::activeMembers = 0;

void Member::assignPlan()
{
    cout << "Enter Member ID: ";
    cin >> memberID;
    cin.ignore();
    cout << "Enter Member Name: ";
    getline(cin, name);
    cout << "Enter Plan (Basic/Premium/VIP): ";
    getline(cin, planType);
    if(planType != "Basic" && planType != "Premium" && planType != "VIP")
    {
        cout << "Invalid Plan! Basic Plan Assigned.\n";
    }
}
```

```

        planType = "Basic";
    }
    activeMembers++;
}

void Member::calculateFee()
{
    if(planType == "Basic")
        fee = 1000;
    else if(planType == "Premium")
        fee = 2000;
    else
        fee = 3000;
}

void Member::display()
{
    cout << "\nMember ID: " << memberID;
    cout << "\nName: " << name;
    cout << "\nPlan Type: " << planType;
    cout << "\nFee: " << fee << endl;
}

int main()
{
    int n;
    cout << "Enter Number of Members: ";
    cin >> n;
    Member m[20];
    for(int i = 0; i < n; i++)
    {
        m[i].assignPlan();
        m[i].calculateFee();
    }
    cout << "\nMembership Details\n";

```

```

for(int i = 0; i < n; i++)

    m[i].display();

Member::showActiveMembers();

return 0;

}

```

The screenshot shows a C++ IDE with a file named 'main.cpp'. The code defines a 'Member' class with attributes 'memberID', 'name', 'planType', and 'fee'. It includes static variables 'activeMembers' and methods for assigning plans, calculating fees, displaying details, and showing active members. The main function prompts the user to enter the number of members, then for each member, their ID, name, and plan type. It assigns a plan (defaulting to 'Basic' if invalid), calculates the fee (1000 for Basic), and displays the details. Finally, it shows the total number of active members.

```

main.cpp
1 #include <iostream>
2 #include <string>
3 using namespace std;
4 class Member
5 {
6 private:
7     int memberID;
8     string name, planType;
9     float fee;
10 public:
11     static int activeMembers;
12     void assignPlan();
13     void calculateFee();
14     void display();
15     static void showActiveMembers()
16     {
17         cout << "\nTotal Active Members: " << activeMembers;
18     }
19 };
20 int Member::activeMembers = 0;
21 void Member::assignPlan()
22 {
23     cout << "Enter Member ID: ";
24     cin >> memberID;
25     cin.ignore();
26     cout << "Enter Member Name: ";

```

Output

```

Enter Number of Members: 2
Enter Member ID: 101
Enter Member Name: meghana
Enter Plan (Basic/Premium/VIP): premium
Invalid Plan! Basic Plan Assigned.
Enter Member ID: 103
Enter Member Name: neha
Enter Plan (Basic/Premium/VIP): basic
Invalid Plan! Basic Plan Assigned.

Membership Details

Member ID: 101
Name: meghana
Plan Type: Basic
Fee: 1000

Member ID: 103
Name: neha
Plan Type: Basic
Fee: 1000

Total Active Members: 2

=== Code Execution Successful ===

```

## Loan Repayment System

```

#include <iostream>

using namespace std;

inline float calculateInterest(float loan, float rate, int years)
{
    return (loan * rate * years) / 100;
}

float calculateEMI(float loan, int years, float rate = 8.5);

void updatePayable(float &amount, float subsidy);

float calculateEMI(float loan, int years, float rate)
{
    return loan + calculateInterest(loan, rate, years);
}

void updatePayable(float &amount, float subsidy)
{
    amount = amount - subsidy;
}

```

```

}
int main()
{
    float loan, rate, subsidy, interest, total;
    int years;
    cout << "Enter Loan Amount: ";
    cin >> loan;
    cout << "Enter Loan Period (Years): ";
    cin >> years;
    cout << "Enter Interest Rate (0 for Default): ";
    cin >> rate;
    if(rate == 0)
        total = calculateEMI(loan, years);
    else
        total = calculateEMI(loan, years, rate);
    if(rate == 0)
        rate = 8.5;
    interest = calculateInterest(loan, rate, years);
    cout << "Enter Subsidy Amount: ";
    cin >> subsidy;
    updatePayable(total, subsidy);
    cout << "\nLoan Amount: " << loan;
    cout << "\nInterest: " << interest;
    cout << "\nTotal Repayment: " << total;
    return 0;
}

```

```
main.cpp  [ ] [ ] [ ] Share Run Output
1 #include <iostream>
2 using namespace std;
3
4 inline float calculateInterest(float loan, float rate, int
  years)
5 {
6     return (loan * rate * years) / 100;
7 }
8
9 float calculateEMI(float loan, int years, float rate = 8.5);
10 void updatePayable(float &amount, float subsidy);
11
12 float calculateEMI(float loan, int years, float rate)
13 {
14     return loan + calculateInterest(loan, rate, years);
15 }
16
17 void updatePayable(float &amount, float subsidy)
18 {
19     amount = amount - subsidy;
20 }
21
22 int main()
23 {
24     float loan, rate, subsidy, interest, total;
25     int years;
```

Enter Loan Amount: 100000  
Enter Loan Period (Years): 2  
Enter Interest Rate (0 for Default): 0  
Enter Subsidy Amount: 5000

Loan Amount: 100000  
Interest: 17000  
Total Repayment: 112000

=== Code Execution Successful ===

## Parcel Charge Management System

```
#include <iostream>

#include <string>

using namespace std;

class Parcel
{
private:
    string type;

    float weight, distance, packagingCharge;

public:
    static int totalParcels;

    void getData();

    float calculateCharge(float packaging = 50);

    float calculateCharge(float packaging, int express);

    void display();

    inline float baseCharge()
    {
        return weight * 10 + distance * 5;
    }

    static void showTotal()
    {
        cout << "\nTotal Parcels Processed: " << totalParcels;
```

```

    }
};

int Parcel::totalParcels = 0;

void Parcel::getData()
{
    cout << "Enter Parcel Type (Normal/Express): ";
    cin >> type;
    cout << "Enter Weight (kg): ";
    cin >> weight;
    cout << "Enter Distance (km): ";
    cin >> distance;
    totalParcels++;
}

float Parcel::calculateCharge(float packaging)
{
    packagingCharge = packaging;
    return baseCharge() + packagingCharge;
}

float Parcel::calculateCharge(float packaging, int express)
{
    packagingCharge = packaging;
    return baseCharge() + packagingCharge + 100;
}

void Parcel::display()
{
    float charge;
    if(type == "Normal")
        charge = calculateCharge();
    else
        charge = calculateCharge(50, 1);
    cout << "\nParcel Type: " << type;
    cout << "\nWeight: " << weight << " kg";
}

```



```

    cout << "\nDistance: " << distance << " km";

    cout << "\nPackaging Charge: " << packagingCharge;

    cout << "\nDelivery Charge: " << charge << endl;
}

int main()
{
    int n;

    cout << "Enter Number of Parcels: ";

    cin >> n;

    Parcel p[20];

    for(int i = 0; i < n; i++)
    {
        p[i].getData();
    }

    for(int i = 0; i < n; i++)
    {
        p[i].display();
    }

    Parcel::showTotal();

    return 0;
}

```

The screenshot shows a C++ IDE with a file named `main.cpp`. The code is a program that calculates the total cost of parcels based on their type, weight, and distance. The output window shows the results of the program execution.

```

main.cpp
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 class Parcel
6 {
7 private:
8     string type;
9     float weight, distance, packagingCharge;
10
11 public:
12     static int totalParcels;
13
14     void getData();
15     float calculateCharge(float packaging = 50);
16     float calculateCharge(float packaging, int express);
17     void display();
18
19     inline float baseCharge()
20     {
21         return weight * 10 + distance * 5;
22     }
23
24     static void showTotal()
25     {
26         cout << "\nTotal Parcels Processed: " <<

```

Output

```

Enter Number of Parcels: 2
Enter Parcel Type (Normal/Express): normal
Enter Weight (kg): 5
Enter Distance (km): 20
Enter Parcel Type (Normal/Express): express
Enter Weight (kg): 3
Enter Distance (km): 12

Parcel Type: normal
Weight: 5 kg
Distance: 20 km
Packaging Charge: 50
Delivery Charge: 300

Parcel Type: express
Weight: 3 kg
Distance: 12 km
Packaging Charge: 50
Delivery Charge: 240

Total Parcels Processed: 2

=== Code Execution Successful ===

```

### **CODE 5: Program: Water Billing System**

```
#include <iostream>
#include <string>
using namespace std;
class Consumer
{
private:
    int id;
    string type;
    float consumption, bill;
public:
    static int totalConsumers;
    void input();
    float calculateBill(float tax = 50);
    float calculateBill(float tax, int commercial);
    void applySubsidy(float &amount);
    void display();
    static void showTotal();
};
int Consumer::totalConsumers = 0;
void Consumer::input()
{
    cout << "Enter Consumer ID: ";
    cin >> id;
    cout << "Enter Consumer Type (Domestic/Commercial): ";
    cin >> type;
    cout << "Enter Water Consumption (Liters): ";
    cin >> consumption;
    totalConsumers++;
}
float Consumer::calculateBill(float tax)
{

```

```

        bill = consumption * 2 + tax;

        return bill;
    }

float Consumer::calculateBill(float tax, int commercial)
{
    bill = consumption * 5 + tax;

    return bill;
}

void Consumer::applySubsidy(float &amount)
{
    float subsidy;

    cout << "Enter Subsidy Amount: ";

    cin >> subsidy;

    amount = amount - subsidy;
}

void Consumer::display()
{
    if(type == "Domestic")
        bill = calculateBill();
    else
        bill = calculateBill(50, 1);

    applySubsidy(bill);

    cout << "\nConsumer ID: " << id;

    cout << "\nConsumer Type: " << type;

    cout << "\nWater Consumption: " << consumption << " Liters";

    cout << "\nBill Amount: " << bill << endl;
}

void Consumer::showTotal()
{
    cout << "\nTotal Consumers: " << totalConsumers;
}

```

```

int main()
{
    int n;

    cout << "Enter Number of Consumers: ";

    cin >> n;

    Consumer c[20];

    for(int i = 0; i < n; i++)

        c[i].input();

    for(int i = 0; i < n; i++)

        c[i].display();

    Consumer::showTotal();

    return 0;
}

```

The screenshot shows a C++ IDE with a file named `main.cpp`. The code defines a `Consumer` array and processes input for 2 consumers. The output window shows the following sequence of events:

```

Enter Number of Consumers: 2
Enter Consumer ID: 101
Enter Consumer Type (Domestic/Commercial): domestic
Enter Water Consumption (Liters): 500
Enter Consumer ID: 100
Enter Consumer Type (Domestic/Commercial): commercial
Enter Water Consumption (Liters): 600
Enter Subsidy Amount: 200

Consumer ID: 101
Consumer Type: domestic
Water Consumption: 500 Liters
Bill Amount: 2350
Enter Subsidy Amount:

```

The IDE interface includes a toolbar with icons for file operations, a 'Share' button, a 'Run' button, and an 'Output' window. The status bar at the bottom shows the temperature as 36°C.